38.2: Flowering plants reproduce sexually, asexually, or both

During **asexual reproduction**, offspring are derived from a single parent without any fusion of egg and sperm. The result is a clone, an individual genetically identical to its parent. Asexual reproduction is common in angiosperms, as well as in other plants, and for some species it is the main mode of reproduction.

Mechanisms of Asexual Reproduction

Asexual reproduction in plants is typically an extension of the capacity for indeterminate growth. Plant growth can be sustained or renewed indefinitely by meristems, regions of undifferentiated, dividing cells (see Concept 35.2). In addition, parenchyma cells throughout the plant can divide and differentiate into more specialized types of cells, enabling plants to regenerate lost parts. Detached root or stem fragments of some plants can develop into whole offspring; for example, pieces of a potato with an “eye” (bud) can each regenerate a whole plant. Such **fragmentation**, the separation of a parent plant into parts that develop into whole plants, is one of the most common modes of asexual reproduction. The plantlets on *Kalanchoë* leaves exemplify an unusual type of fragmentation (see Figure 35.7). In other cases, the root system of a single parent, such as an aspen tree, can give rise to many shoots that become separate shoot systems **(Figure 38.13)**. One aspen clone in Utah has been estimated to be composed of 47,000 stems of genetically identical trees. Although it is likely that some of the root system connections have been severed, making some of the trees isolated from the rest of the clone, each tree still shares a common genome.

Figure 38.13

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Asexual reproduction in aspen trees.
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Some aspen groves, such as those shown here, consist of thousands of trees descended by asexual reproduction. Each grove of trees derives from the root system of one parent. Thus, the grove is a clone. Notice that genetic differences between groves descended from different parents result in different timing for the development of fall color.

A different mechanism of asexual reproduction has evolved in dandelions and some other plants. These plants can sometimes produce seeds without pollination or fertilization. This asexual production of seeds is called **apomixis** (from the Greek words meaning “away from the act of mixing”) because there is no joining or, indeed, production of sperm and egg. Instead, a diploid cell in the ovule gives rise to the embryo, and the ovules mature into seeds, which in the dandelion are dispersed by windblown fruits. Thus, these plants clone themselves by an asexual process but have the advantage of seed dispersal, usually associated with sexual reproduction. Plant breeders are interested in introducing apomixis into hybrid crops because it would allow hybrid plants to pass desirable genomes intact to offspring.

Advantages and Disadvantages of Asexual and Sexual Reproduction

EVOLUTION An advantage of asexual reproduction in some cases is that there is no need for pollination. This may be beneficial in situations where plants of the same species are sparsely distributed and unlikely to be successfully pollinated. Asexual reproduction also allows the plant to pass on all its genetic legacy intact to its progeny. In contrast, when reproducing sexually, a plant passes on only half of its alleles. If a plant is superbly suited to its environment, asexual reproduction can be advantageous. A vigorous plant can potentially clone many copies of itself, and if the environmental circumstances remain stable, these offspring will also be genetically well adapted to the same environmental conditions under which the parent flourished.

Asexual plant reproduction based on the vegetative growth of stems, leaves, or roots is known as **vegetative reproduction**. Generally, the progeny produced by vegetative reproduction are stronger than seedlings produced by sexual reproduction. In contrast, seed germination is a precarious stage in a plant's life. The tough seed gives rise to a fragile seedling that may face exposure to predators, parasites, wind, and other hazards. In the wild, few seedlings survive to become parents themselves. Production of enormous numbers of seeds compensates for the odds against individual survival and gives natural selection ample genetic variations to screen. However, this is an expensive means of reproduction in terms of the resources consumed in flowering and fruiting.

Because sexual reproduction generates variation in offspring and populations, it can be advantageous in unstable environments where evolving pathogens and other fluctuating conditions affect survival and reproductive success. In contrast, the genotypic uniformity of asexually produced plants puts them at great risk of local extinction if there is a catastrophic environmental change, such as a new strain of disease. Moreover, seeds (which are almost always produced sexually) facilitate the dispersal of offspring to more distant locations. Finally, seed dormancy allows growth to be suspended until environmental conditions become more favorable. In the **Scientific Skills Exercise**, you can use data to determine which species of monkey flower are mainly asexual reproducers and which are mainly sexual reproducers.

Scientific Skills Exercise

Using Positive and Negative Correlations to Interpret Data

AP Science Practices 2, 4, 5, 6

Do Monkey Flower Species Differ in Allocating Their Energy to Sexual Versus Asexual Reproduction? Over the course of its life span, a plant captures only a finite amount of resources and energy, which must be allocated to best meet the plant's individual requirements for maintenance, growth, defense, and reproduction. Researchers examined how five species of monkey flower (genus *Mimulus*) use their resources for sexual and asexual reproduction.

How the Experiment Was Done After growing specimens of each species in separate pots in the open, the researchers determined averages for nectar volume, nectar concentration, seeds produced per flower, and the number of times the plants were visited by broad-tailed hummingbirds (*Selasphorus platycercus*, shown here). Using greenhouse-grown specimens, they determined the average number of rooted branches per gram fresh shoot weight for each of the species. The phrase *rooted branches* refers to asexual reproduction through horizontal shoots that develop roots.

Data from the Experiment

Species	Nectar Volume (μL)	Nectar Concentration (% weight of sucrose/ total weight)	Seeds per Flower	Visits per Flower	Rooted Branches per Gram Shoot Weight
M. rupestris	4.93	16.6	2.2	0.22	0.673
M. eastwoodiae	4.94	19.8	25	0.74	0.488
M. nelsonii	20.25	17.1	102.5	1.08	0.139
M. verbenaceus	38.96	16.9	155.1	1.26	0.091
M. cardinalis	50.00	19.9	283.7	1.75	0.069

Data from S. Sutherland and R. K. Vickery, Jr., Trade-offs between sexual and asexual reproduction in the genus *Mimulus*, *Oecologia* 76:330–335 (1998).

Species	Nectar volume (microliter)	Nectar Concentration (percent weight of sucrose divided by total weight)	Seeds per flower	Visits per flower	Rooted Branches per Gram Shoot Weight
M. rupestris	4.93	16.6	2.2	0.22	0.673
M. eastwoodiae	4.94	19.8	25	0.74	0.488
M. nelsonii	20.25	17.1	102.5	1.08	0.139
M. verbenaceus	38.96	16.9	155.1	1.26	0.091
M. cardinalis	50.00	19.9	283.7	1.75	0.069

INTERPRET THE DATA

- A correlation is a way to describe the relationship between two variables. In a positive correlation, as the values of one of the variables increase, the values of the second variable also increase. In a negative correlation, as the values of one of the variables increase, the values of the second variable decrease. Or there may be no correlation between two variables. If researchers know how two variables are correlated, they can make a prediction about one variable based on what they know about the other variable.
- Which variable(s) is/are positively correlated with the volume of nectar production in this genus?
- Which is/are negatively correlated?
- Which show(s) no clear relationship?
-
    - Which *Mimulus* species would you categorize as mainly asexual reproducers? Why?
- Which species would you categorize as mainly sexual reproducers? Why?
-
    - Which species would probably fare better in response to a pathogen that infects all *Mimulus* species?
- Which species would fare better if a pathogen caused hummingbird populations to dwindle?

Instructors: A version of this Scientific Skills Exercise can be assigned in **Mastering Biology**.

Although sexual reproduction involving two genetically different plants produces the most genetically diverse offspring, some plants, such as garden peas, usually self-fertilize. This process, called “selfing,” is a desirable attribute in some crop plants because it heightens the probability that every ovule will develop into a seed. In many angiosperm species, however, mechanisms have evolved that make it difficult or impossible for a flower to fertilize itself, as we’ll see next.

Mechanisms That Prevent Self-Fertilization

The various mechanisms that prevent self-fertilization contribute to genetic variety by ensuring that the sperm and egg come from different parents. Some species of plants cannot self-fertilize because different individuals have either staminate flowers (lacking carpels) or carpellate flowers (lacking stamens) **(Figure 38.14a)**. Other plants have flowers with functional stamens and carpels that mature at different times or are structurally arranged in such a way that it is unlikely that an animal pollinator could transfer pollen from an anther to a stigma of the same flower **(Figure 38.14b)**. However, the most common anti-selfing mechanism in flowering plants is **self-incompatibility**, the ability of a plant to reject its own pollen and the pollen of closely related individuals. If a pollen grain lands on a stigma of a flower of the same plant or a closely related plant, a biochemical block prevents the pollen from completing its development and fertilizing an egg. This plant response is analogous to the immune response of animals because both are based on the ability to distinguish the cells of “self” from those of “nonself.” The key difference is that the animal immune system rejects nonself, as when the immune system mounts a defense against a pathogen or rejects a transplanted organ (see Concept 43.3). In contrast, self-incompatibility in plants is a rejection of self.

Figure 38.14

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Some floral adaptations that prevent self-fertilization.
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(a) Some species, such as *Sagittaria latifolia* (common arrowhead), have plants that produce only staminate flowers (left) or carpellate flowers (right).

(b) Some species, such as *Oxalis alpina* (alpine wood sorrel), produce two types of flowers on different individuals: "thrums," which have short styles and long stamens, and "pins," which have long styles and short stamens. An insect foraging for nectar would collect pollen on different parts of its body; thrum pollen would be deposited on pin stigmas, and vice versa.

Part a. Some species, such as Sagittaria latifolia (common arrowhead), are dioecious, having plants that produce only staminate flowers (left) or carpellate flowers (right).

Part b. Some species, such as Oxalis alpina (alpine wood sorrel), produce two types of flowers on different individuals: thrums, which have short styles and long stamens, and pins, which have long styles and short stamens. An insect foraging for nectar would collect pollen on different parts of its body; thrum pollen would

be deposited on pin stigmas, and vice versa.

Researchers are unraveling the molecular mechanisms of self-incompatibility. Recognition of “self” pollen is based on genes called S-genes. In the gene pool of a population, there can be dozens of alleles of an S-gene. If a pollen grain has an allele that matches an allele of the stigma on which it lands, the pollen tube either fails to germinate or fails to grow through the style to the ovary. There are two types of self-incompatibility: gametophytic and sporophytic.

In gametophytic self-incompatibility, the S-allele in the pollen genome governs the blocking of fertilization. For example, an S1 pollen grain from an S1S2 parental sporophyte cannot fertilize eggs of an S1S2 flower but can fertilize an S2S3 flower. An S2 pollen grain cannot fertilize either flower. In some plants, this self-recognition involves the enzymatic destruction of RNA within a pollen tube. RNA-hydrolyzing enzymes are produced by the style and enter the pollen tube. If the pollen tube is a “self” type, they destroy its RNA.

In sporophytic self-incompatibility, fertilization is blocked by incompatibilities between S-allele gene products present in the sporophytic parental tissue adhering to the pollen wall and S-allele gene products secreted by the stigma of the receptive flower. For example, both the S1 and S2 pollen grains derived from an S1S2 parental sporophyte contain S1-allele and S2-allele gene products in the sporophytic tissue adhering to their pollen walls. The S2-allele gene product in the pollen walls of these two pollen types would prevent either type from germinating on the stigma of a flower whose genotype includes either the S1 or the S2 allele. For example, an S1 pollen grain, derived from an S1S2 parent, cannot germinate on the stigma of an S2S3 flower.

Research on self-incompatibility may have agricultural applications. Breeders often hybridize different genetic strains of a crop to combine the best traits of the two strains and to counter the loss of vigor that can result from excessive inbreeding. To prevent self-fertilization within the two strains, breeders must either laboriously remove the anthers from the parent plants that provide the seeds or use male-sterile strains of the crop plant, if they exist. If self-compatibility can be genetically engineered back into domesticated plant varieties, these limitations to the commercial hybridization of crop seeds could be overcome.

Totipotency, Vegetative Reproduction, and Tissue Culture

In a multicellular organism, any cell that can divide and asexually generate a clone of the original organism is said to be **totipotent**. Totipotency is found in many plants, particularly but not exclusively in their meristematic tissues. Plant totipotency underlies most of the techniques used by humans to clone plants.

Vegetative Propagation and Grafting

Vegetative reproduction occurs naturally in many plants, but it can often be facilitated or induced by humans, in which case it is called **vegetative propagation**. Most houseplants, landscape shrubs and bushes, and orchard trees are asexually reproduced from plant fragments called cuttings. In most cases, shoot cuttings are used. At the wounded end of the shoot, a mass of dividing, undifferentiated totipotent cells called a **callus** forms, and roots develop from the callus. If the shoot fragment includes a node, then roots form without a callus stage.

In grafting, a severed shoot from one plant is permanently joined to the truncated stem of another. This process, usually limited to closely related individuals, can combine the best qualities of different species or varieties into one plant. The plant that provides the roots is called the **stock**; the twig grafted onto the stock is known as the **scion**. For example, scions from varieties of vines that produce superior wine grapes are grafted onto rootstocks of varieties that produce inferior grapes but are more resistant to certain soil pathogens. The genes of the scion determine the quality of the fruit. During grafting, a callus first forms between the adjoining cut ends of the scion and stock; cell differentiation then completes the functional unification of the grafted individuals.

Test-Tube Cloning and Related Techniques

Plant biologists have adopted *in vitro* methods to clone plants for research or horticulture. Whole plants can be obtained by culturing small pieces of tissue from the parent plant on an artificial medium containing nutrients and hormones. The cells or tissues can come from any part of a plant, but growth may vary depending on the plant part, species, and artificial medium. In some media, the cultured cells divide and form a callus of undifferentiated totipotent cells **(Figure 38.15a)**. When the concentrations of hormones and nutrients are manipulated appropriately, a callus can sprout shoots and roots with fully differentiated cells **(Figure 38.15bandc)**. If desired, the cloned plantlets can then be transferred to soil, where they continue their growth.

Figure 38.15

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Cloning a garlic plant.
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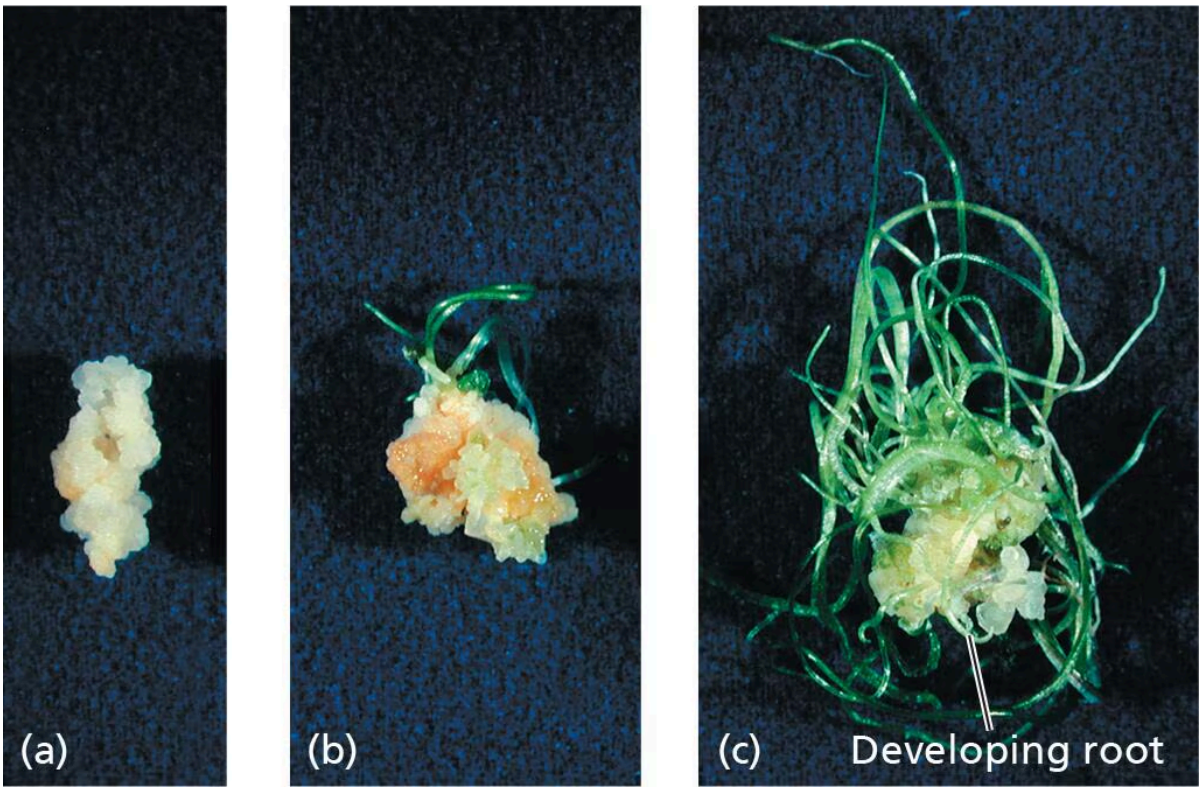

(a) *A root from a garlic clove gave rise to this callus culture, a mass of undifferentiated totipotent cells.* ***(b and c)*** *The differentiation of a callus into a plantlet depends on the nutrient levels and hormone concentrations in the artificial medium, as can be seen in these cultures grown for different lengths of time.*

Plant tissue culture is important in eliminating weakly pathogenic viruses from vegetatively propagated varieties. Although the presence of weak viruses may not be obvious, yield or quality may be substantially reduced as a result of infection. Strawberry plants, for example, are susceptible to more than 60 viruses, and typically the plants must be replaced each year because of viral infection. However, since the apical meristems are often virus-free, they can be excised and used to produce virus-free material for tissue culture.

Plant tissue culture also facilitates genetic engineering, the deliberate modification of the characteristics of an organism by manipulating its genetic material (see Concept 20.1). Many genetic engineering techniques require single plant cells as the starting material. Test-tube culture makes it possible to regenerate plants from a single genetically engineered plant cell. In the next section, we'll take a closer look at some of the promises and challenges surrounding the use of genetically modified plants in agriculture.